

DeepCue

Real-Time Multimodal Emotion Recognition
for Hebrew-Speaking Interview Candidates

“What a candidate says, how they say it, and what their face shows — fused by a Transformer.”



Rachel Brodsky · Solo Project · Deep Learning Final Project

The Problem: Emotion Is Multimodal

Any single channel is ambiguous — especially when candidates actively regulate their expression



Facial micro-expressions

Brief, involuntary movements — a calm voice can mask an anxious face.



Vocal prosody

Pitch, energy, tempo — positive words can ride on strained, sarcastic delivery.



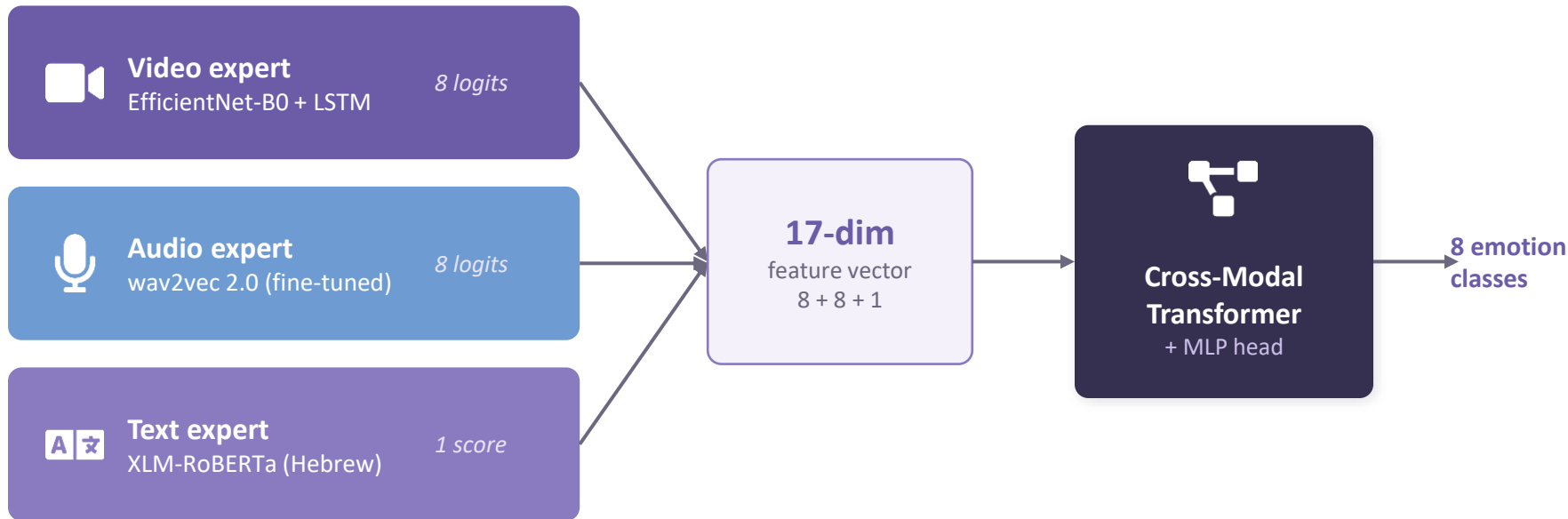
Word semantics

What is actually said — and almost no emotion research targets Hebrew.

The interview setting raises the stakes: candidates self-regulate, producing subtle and conflicting cues — exactly the cases where a single-modality model fails.

The Solution: A Mixture of Experts

Three independently trained deep models, fused by a Cross-Modal Transformer



neutral · confident · anxious · happy · sad · angry · surprised · uncertain

Design principle: decouple the experts — each can be retrained or upgraded without touching the others.

Design Constraints That Shaped Every Choice

Deployment reality drove the deep learning decisions, not the other way around



CPU-only inference

Weak Windows machine, no GPU — every model exports to ONNX and must run fast.



< 10 s end-to-end latency

From raw signal to fused prediction, live during the interview.



Quality gate: macro F1 ≥ 0.50

Per classifier, on strictly held-out data.



Compact by necessity

EfficientNet-B0, 3-second audio chunks, landmarks instead of raw pixels.

The Data

Two public datasets — and the evaluation hygiene that made results trustworthy



RAVDESS — video & audio experts

- 24 professional actors, 8 emotion categories, audio-visual recordings.
- Actor-disjoint splits: no actor in both train and validation — prevents identity leakage.
- *An early leaked split scored ~ 0.99 F1 — too good to be true, and it was.*



Hebrew Sentiment — text expert

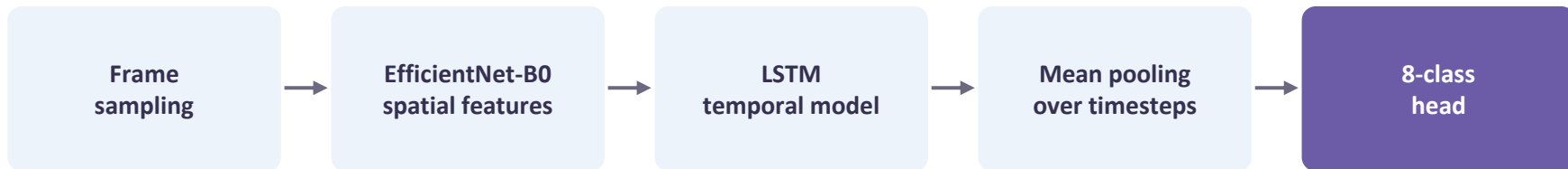
- `omilab/hebrew_sentiment` corpus.
- A pragmatic pivot with a bonus: the text expert became natively Hebrew.
- Continuous sentiment labels → regression task.



Stated honestly: RAVDESS has no “confident” category, so that class has zero support in video/audio evaluation.

Video Expert: EfficientNet-B0 + LSTM

Spatial features per frame, temporal dynamics across frames



- Transfer learning from ImageNet weights; horizontal-flip augmentation; actor-disjoint validation.
- Mean pooling over LSTM timesteps outperformed last-timestep readout.
- Why this pair: B0 is the best accuracy-per-FLOP fit for CPU deployment; the LSTM captures micro-expression dynamics a single frame misses.

0.80

macro F1 · 8 classes
actor-disjoint evaluation

Audio Expert: Fine-Tuned wav2vec 2.0

Paralinguistics, not words — the model hears how something is said

1

Stage 1 — head only

Backbone frozen; train the 8-class classification head on RAVDESS audio.

2

Stage 2 — full fine-tune

Unfreeze the Transformer; best validation F1 $\approx$ 0.73 on the audio-only set.

- Self-supervised speech Transformer pre-trained on raw waveforms — reads prosody, energy, and voice quality directly.
- Speech is transcribed to text in real time using OpenAI's Whisper model, which feeds the extracted content into the semantic/text emotion pipeline alongside the paralinguistic audio features.

0.45

macro F1 · cross-source
evaluation

*Trained on audio-only WAVs, evaluated
on audio from AV MP4s — a domain-
shift lesson, revisited in Results.*

Text Expert: XLM-RoBERTa for Hebrew Sentiment

A multilingual Transformer, fine-tuned as a regressor

- Why XLM-R: the strongest open multilingual encoder (100 languages) for a language that is low-resource in sentiment analysis.
- Why regression: sentiment intensity is a graded signal — a continuous score feeds fusion better than a hard label.
- Evaluated with MAE — the correct metric for a regression head.
- *Future work: upgrade to an 8-class head for architectural symmetry with the sibling experts.*

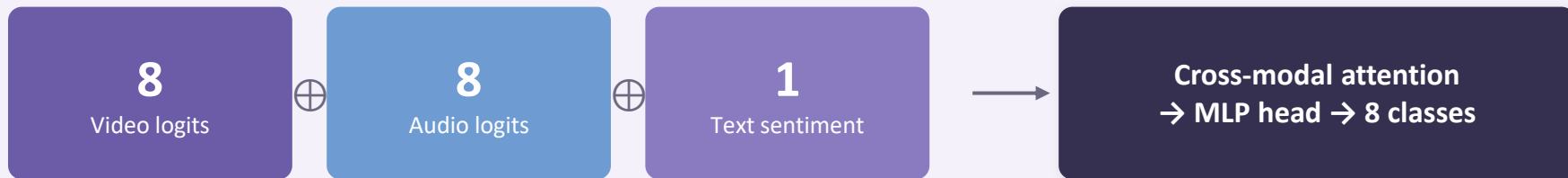
0.046

MAE · normalized scale

*predictions land within ~5%
of ground truth on average*

The Cross-Modal Transformer

Learned, context-dependent weighting — not fixed averaging



Each modality's evidence re-weights the others.

A confident voice can override an ambiguous face in one moment; the face can dominate in the next. That context-dependent weighting is exactly what conflicting-cue cases require — and what fixed-weight averaging or majority voting cannot do.

The Architectural Viability Simulation

Proving the fusion architecture before real paired data exists

The core challenge: no paired multimodal dataset exists for Hebrew interview emotion — the same moment would need labels across all three channels at once.

The deliberate strategy: construct a synthetic dataset simulating the empirical output distributions of the three trained experts (intensity-modulated class means, realistic overlap) — an architectural proof-of-concept that de-risks the pipeline before real paired data arrives.

0.41 → 0.98

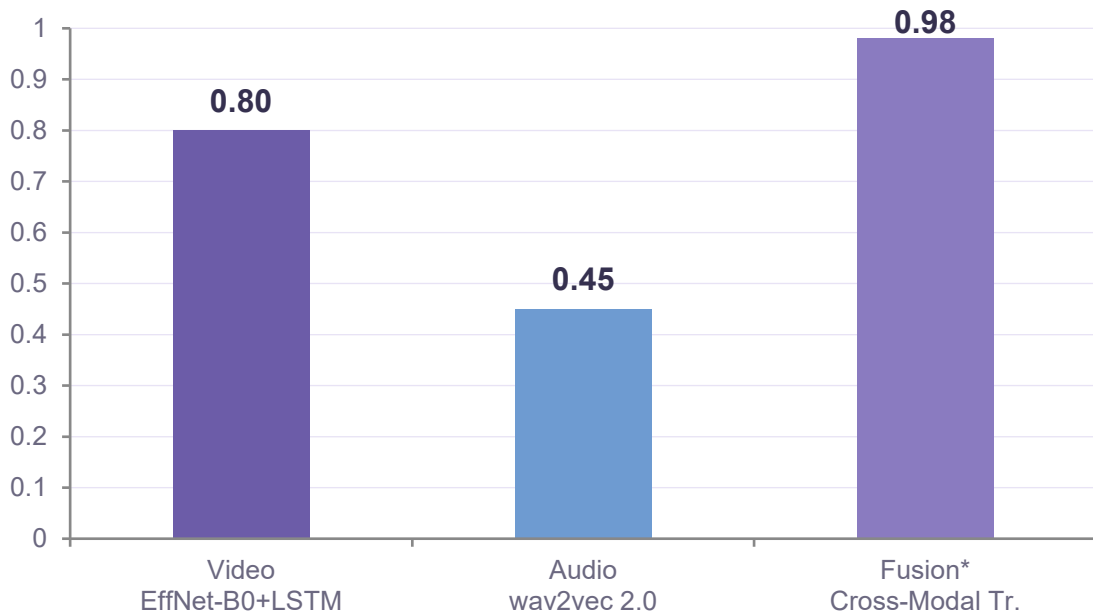
macro F1, first vs. refined simulation

The first simulation used overlapping class means and scored 0.41. Diagnosing that as a data problem — not a model failure — is itself a deep learning lesson.

The 0.98 validates the attention mechanism's capacity on the 17-dim space — real-world fusion performance awaits paired data from deployed sessions.

Results & Evaluation

Macro F1 = per-class fairness under imbalance · MAE = average error for regression



quality gate: macro F1 ≥ 0.50 — audio sits just below

**Fusion evaluated on the Architectural Viability Simulation (synthetic paired data).*

Text: MAE = 0.046

Regression head — ~5% average error on the normalized sentiment scale.

Audio: 0.45

Just under the gate — explained by cross-source domain shift; an open fine-tuning item.

Challenges & Engineering Decisions

The honest numbers came from finding what was wrong with the flattering ones



Quantization dropped for ONNX stability

INT8 shrank the video model 74% (33.3 → 8.7 MB) but destabilized the ONNX inference path. Shipped full-precision — still within the CPU latency budget. Compression is only free on paper.



Actor leakage

The first video result (~0.99 F1) was too good to be true. Actor-disjoint splits revealed the honest number: 0.80.



Metric-model matching

The text expert was initially gated with F1 despite being a regressor — corrected to MAE, the right metric for the task.



Domain shift in audio

Train/eval source mismatch (WAV-only vs. AV-extracted audio) is the leading suspect for the 0.45 — documented, not hidden.

Future Work

The simulation was the rehearsal — real paired data is the performance



Real paired multimodal data Deployed interview sessions generate genuinely paired video-audio-text moments → retrain fusion on real distributions.



Audio fine-tuning pass Fine-tune on AV-sourced audio to close the domain gap behind the 0.45.



Text: regression → classification Upgrade to an 8-class head for architectural symmetry across all three experts.



Quantization, revisited Quantization-aware training (QAT) instead of post-training INT8 — keep the size win without the instability.

Thank you!

github.com/RachelBSade/DeepCue